ECON102 - Assignment 1

Xiao Lu • 2026-10-12

---

1. **ANS**:

   Previously we have $C = 0.75Y$, now we have $C = 0.7Y$.

   Let Y_I be initial aggregate income and Y_N be the new aggregate income

   $$Y_I = C_I + I = 0.75Y_I + I = 0.75Y_I + 40 + 0.05Y_I = 0.8Y_I + 40 \Longrightarrow Y_I = 200$$

   $$Y_N = C_N + I = 0.7Y_N + I = 0.7Y_N + 40 + 0.05Y_N = 0.75Y_N + 40 \Longrightarrow Y_N = 160$$

   $$S_I = Y_I - C_I = 200 - 150 = 50$$

   $$S_N = Y_N - C_N = 160 - 112 = 48$$

   $S_I = 50 > S_N = 48$. The attempt of increasing aggregate saving failed. When households attempt to save more by reducing consumption, aggregate spending falls. This lowers firms' revenues and equilibrium aggregate income. The fall in income is large enough that aggregate saving falls from 50 to 48. This is called the paradox of thrift.

2. **ANS**:

   a)

   Smelter: $90000 - 30000 = 60000$

   Bicycle maker: $60000 + 150000 - 90000 - 50000 = 70000$

   Retailer: $220000 - 150000 = 70000$

   Total GDP = $60000 + 70000 + 70000 = 200000$

   b)

   Smelter: $90000 - 30000 = 60000$

   Supplier: $50000 - 0 = 50000$

   Bicycle maker: $60000 + 150000 - 90000 - 50000 = 70000$

   Retailer: $220000 - 150000 = 70000$

   Total GDP = $60000 + 70000 + 70000 + 50000 = 250000$

   The GDP increases by 50000 because there is more goods produce within Canada that increases Canadian GDP rather than from foreign import.